

Yunfei Tian

Tel: (+86)187 5200 9698 E-mail: tyfwin@163.com

Educational background

2013.09-2017.06 Shandong University (Undergraduate) Optoelectronic Engineering

- GPA: 4.35/5.0; Rank: 3/42; Awarded national encouragement scholarship.
- Won 2 international honors, 5 national honors, and 6 provincial honors.

2017.09-2020.06 Nanjing University (Master) Optical Engineering

- GPA: 4.5/5.0; Rank: 2/43; Awarded national scholarship.
- Research direction: computational imaging.
- Published 3 SCI/EI papers and obtained 6 authorized patents.

Work experience

2020.07-Present Huawei Technologies Co., Ltd. Algorithm Engineer

- Experience in developing, integrating, and implementing image algorithms for classification, detection, segmentation, and tracking in areas such as mobile phone cameras and intelligent manufacturing.
- Familiarity with optical imaging, computer vision, intelligent manufacturing, camera and image signal processing, 3A technologies, and experience in developing and deploying applications on both cloud and edge platforms.
- Work honors include "Tomorrow's Star," "Overcoming Difficulties Award," and "Excellent Team Award" for the Mate50 phone project.

Project experience

2022.02-2023.05 Object detection and tracking algorithm of Huawei flagship smartphone camera

- As the feature owner, responsible for designing algorithms, model training and quantification, algorithm integration, effect verification, and other related tasks for object detection and tracking in camera preview algorithms.
- Optimize the algorithm to reduce power consumption and load as much as possible, meet the commercial requirements of smartphone CPU and NPU resources.
- Cooperate with upstream and downstream teams to complete multi-camera mode switching, AWB, AE, AF, APK frame drawing, and other functions.
- Successfully implemented on Huawei's Mate50 and P60 series phones, supporting Huawei's top selling features such as long focal length moon mode and 4D focus.

2021.05-2022.02 Virtual measurement technology in semiconductor manufacturing

- Reduced the frequency of measurements, increased the utilization of measurement equipment, and improved production capacity in a certain semiconductor manufacturing process.
- Abstracted structured data into images and used image interpolation and super-resolution algorithms for reconstruction. Used Attention mechanisms to help users select important measurement points. Also responsible for algorithm service deployment and maintenance.

- Successfully implemented in foundries, breaking the foreign technology monopoly in the field, and applied for an invention patent.

2020.07-2021.05 Wafermap image classification in semiconductor manufacturing

- Automatic classification of WaferMaps with different patterns can help engineers quickly analyze production status in semiconductor manufacturing.
- Addressed issues such as insufficient data, unlabeled images, annotation errors, multiple labels, and unknown categories using methods such as Autoencoders (AE), Variational Autoencoders (VAE), Transfer Learning, OpenSet Loss, and Center Loss.
- Successfully implemented in foundries, helping engineers locate production faults and establish a smart factory.

2019.06-2019.08 Semantic segmentation algorithm (Huawei Internship)

- Constructed a language dataset using adaptive feathering, erosion and dilation, and Poisson fusion algorithms.
- Trained traditional segmentation algorithms based on superpixel segmentation, while also training lightweight semantic segmentation network models.

Research experience

2017.10-2018.12 Super-resolution microscopy imaging.

- Designed and built a low magnification, large field-of-view optical imaging system for microscopy, using multiple light sources to capture multiple low-resolution, large field-of-view intensity images.
- Implemented the Fourier ptychographic imaging algorithm to improve the resolution by more than 10 times. Combined with deep learning-based super-resolution algorithms, further improved the resolution.

2019.01-2020.03 Fiber-based single-pixel imaging.

- Used different randomly independent structured light to illuminate the object, and detected the corresponding scattered light power by using a detector. Utilized a fiber-optic head as a single-pixel receiver to achieve remote imaging through fiber-optic transmission.
- Used matrix pseudo-inverse algorithm to calculate the reconstructed object pattern based on the scattered light patterns generated by known multiple LED screens and detected by the detector. Combined with compressed sensing algorithm to reduce the number of detection times and data volume, and speed up the imaging process.

Skills

- Proficient in programming languages such as Python, Matlab, and C++.
- Familiar with deep learning frameworks such as PyTorch, TensorFlow, and OpenCV.
- Familiar with tools such as Linux, JMESPath, SQL, K8s, Helm, Git, and Django.
- Proficient in software such as Zemax, Labview, Photoshop, Premiere, Lightroom, and Office.
- Experienced in control system development, including 51, MSP430, STM32, Arduino, and Android phones.
- English: IELTS 6.0.